

Body Fat and Weight Loss Calculator Report - Test User

Report Date: 06/29/2025

1. Current Profile

Height:	6'0" (182.88 cm)
Current Weight:	200.00 lbs
Current Body Fat:	20.00%
Current Lean Mass:	160.00 lbs
Current Fat Mass:	40.00 lbs
Age:	30 years
Gender:	m

2. Target Goals

Goal Weight:	180.00 lbs
Goal Body Fat:	12.00%
Timeline:	06/29/25 to 12/28/25
Total Duration:	26 weeks

3. Current Activity Profile

Activity Level:	Light exercise/sports 1-3 days/week
Training Experience:	Intermediate (2-4 years)
Workout Type:	Bodybuilding
Weekly Workout Frequency:	4 days
Currently Using Resistance Training:	Yes
Athlete Status:	Yes

Understanding Your Activity Profile:

- **Activity Level:** Your Light exercise/sports 1-3 days/week lifestyle contributes significantly to your daily caloric expenditure.
- **Training Experience:** As a Intermediate (2-4 years) trainee, this affects your potential muscle gain and recovery needs.
- **Workout Frequency:** Training 4 days per week helps optimize your metabolic rate and recovery periods.
- **Resistance Training:** Your use of resistance training enhances muscle preservation during weight loss.

4. Current Metabolic Profile

Base Metabolic Rate (RMR):	1900.00 cal/day
Total Daily Energy Expenditure (TDEE):	4091.00 cal/day
Thermic Effect of Food (TEF):	239.40 cal/day
Non-Exercise Activity Thermogenesis (NEAT):	500.00 cal/day
Recommended Daily Calorie Intake:	3721.33 cal/day

Understanding Your Metabolic Profile:

- **RMR (Resting Metabolic Rate):** Your body burns 1900.00 calories daily at rest. This is calculated using the Mifflin-St Jeor equation, adjusted for your body composition.
- **TDEE (Total Daily Energy Expenditure):** Based on your activity level, you burn approximately 4091.00 calories per day.
- **TEF (Thermic Effect of Food):** You burn about 239.40 calories digesting food, particularly protein.
- **NEAT:** Your daily activities burn approximately 500.00 additional calories.

Additional Lifestyle Factors

Target Daily Protein Intake:	180.0 grams
Diet Type:	high_protein
Reported PED Use:	No
Reported Sleep Quality:	good

Impact of These Factors:

- **Protein Intake (180.0g):** Adequate protein is crucial for muscle repair, growth, and satiety. Higher protein intake also increases TEF.
- **Diet Type (high_protein):** Your chosen diet type influences macronutrient distribution, which can affect energy levels, satiety, and adherence. Specific diet types also have varying impacts on TEF and nutrient partitioning.
- **PED Use (No):** If PEDs are used, this can significantly alter metabolic rate, muscle protein synthesis, and nutrient partitioning, leading to faster changes than naturally achievable. Calculations are adjusted accordingly.
- **Sleep Quality (good):** Sleep is vital for recovery, hormone regulation (including those related to appetite and muscle growth), and overall well-being. Poor sleep can hinder progress.

5. Workout Analysis

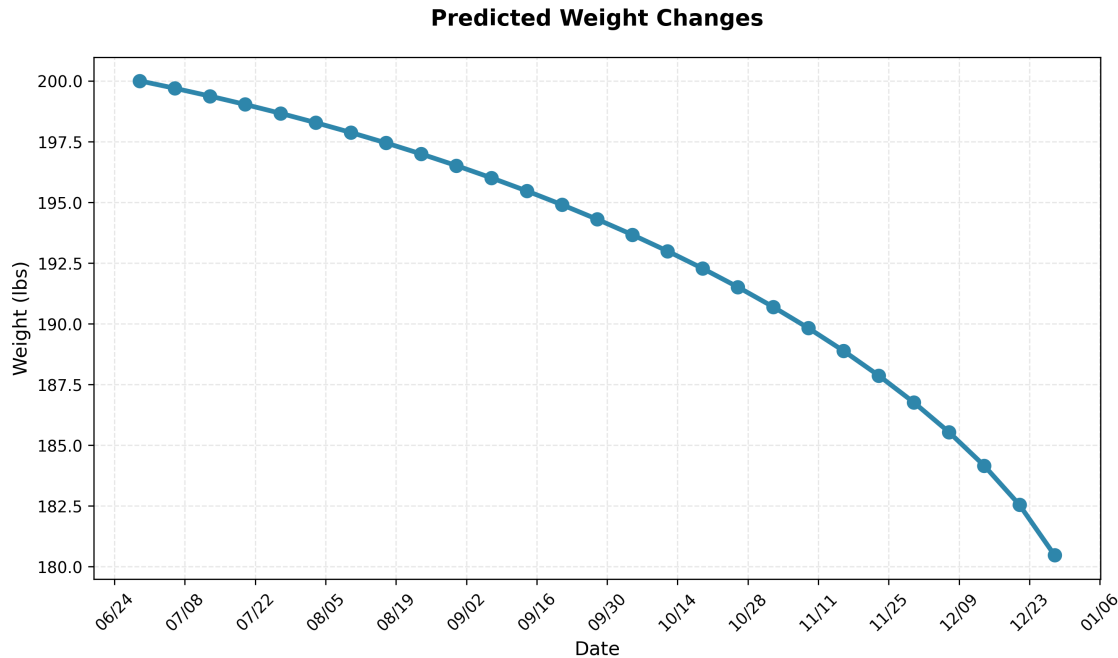
Type:	Bodybuilding
Frequency:	4 days/week
Volume Score:	0.57 - Moderate
Intensity Score:	0.80 - High
Frequency Score:	1.00 - High

Impact on Your Caloric Needs:

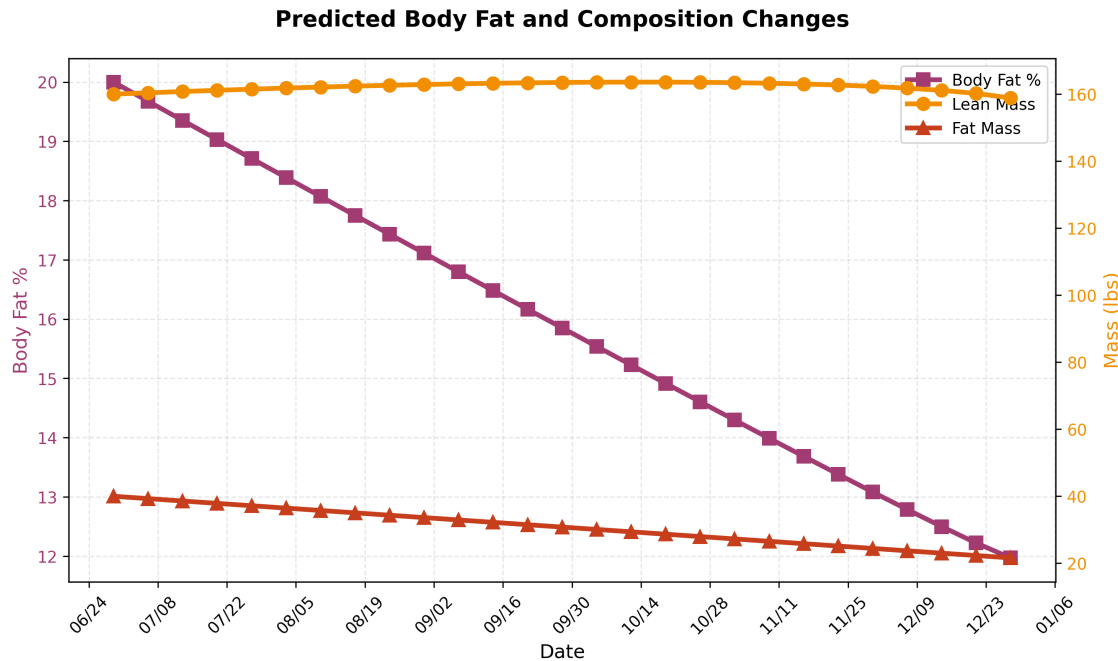
- **Volume Score 0.57:** Your training volume contributes approximately 285.71 additional calories to your daily expenditure.
- **Intensity Score 0.80:** Your training intensity increases your metabolic rate by about 12.00% for up to 48 hours post-workout.
- **Frequency Score 1.00:** Training 4 days per week optimizes your caloric burn and recovery periods.

6. Predicted Progress Charts

Predicted Weight Changes



Predicted Body Fat Changes



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7. Weekly Progress Forecast

Date	Weight (lbs)	Body Fat %	Daily Cal Intake	TDEE	Weekly Cal Output	Total Weight Lost (lbs)	Lean Mass (lbs)	Fat Mass (lbs)	Muscle Gain (lbs)	RMR
06/29/25	200.00	20.00	3721.33	4091.00	0.00	0.00	160.00	40.00	0.00	1900.00
07/06/25	199.69	19.68	3721.33	4091.00	2587.69	0.31	160.40	39.29	0.46	1900.00
07/13/25	199.37	19.35	3714.56	4089.00	2621.10	0.63	160.79	38.58	0.46	1899.00
07/20/25	199.03	19.03	3707.58	4087.00	2655.91	0.97	161.15	37.88	0.46	1897.00
07/27/25	198.66	18.71	3700.40	4085.00	2692.23	1.34	161.50	37.17	0.46	1896.00
08/03/25	198.28	18.39	3691.97	4082.00	2730.20	1.72	161.82	36.46	0.46	1894.00
08/10/25	197.87	18.07	3683.29	4079.00	2769.97	2.13	162.12	35.75	0.46	1892.00
08/17/25	197.44	17.75	3674.32	4076.00	2811.74	2.56	162.40	35.05	0.46	1891.00
08/24/25	196.99	17.43	3665.04	4073.00	2855.70	3.01	162.65	34.34	0.46	1889.00
08/31/25	196.51	17.11	3655.41	4070.00	2902.11	3.49	162.88	33.63	0.46	1887.00
09/07/25	196.00	16.80	3645.39	4067.00	2951.24	4.00	163.08	32.92	0.46	1884.00
09/14/25	195.47	16.48	3633.94	4063.00	3003.45	4.53	163.25	32.22	0.46	1882.00
09/21/25	194.90	16.17	3621.98	4059.00	3059.14	5.10	163.39	31.51	0.46	1880.00
09/28/25	194.30	15.85	3609.46	4055.00	3118.80	5.70	163.50	30.80	0.46	1877.00
10/05/25	193.66	15.54	3596.28	4051.00	3183.06	6.34	163.57	30.09	0.46	1874.00
10/12/25	192.99	15.23	3582.33	4047.00	3252.67	7.01	163.60	29.38	0.46	1871.00
10/19/25	192.27	14.91	3566.48	4042.00	3328.61	7.73	163.60	28.68	0.46	1868.00
10/26/25	191.51	14.60	3549.55	4037.00	3412.14	8.49	163.54	27.97	0.46	1865.00
11/02/25	190.69	14.30	3531.29	4032.00	3504.95	9.31	163.43	27.26	0.46	1862.00
11/09/25	189.82	13.99	3510.38	4026.00	3609.36	10.18	163.27	26.55	0.46	1858.00
11/16/25	188.88	13.68	3487.33	4020.00	3728.69	11.12	163.04	25.85	0.46	1854.00
11/23/25	187.87	13.38	3460.44	4013.00	3867.91	12.13	162.73	25.14	0.46	1850.00
11/30/25	186.76	13.08	3429.57	4006.00	4034.98	13.24	162.33	24.43	0.46	1845.00
12/07/25	185.53	12.79	3391.74	3998.00	4243.80	14.47	161.81	23.72	0.46	1840.00
12/14/25	184.15	12.50	3343.97	3990.00	4522.24	15.85	161.14	23.02	0.46	1835.00
12/21/25	182.54	12.22	3274.30	3980.00	4939.90	17.46	160.23	22.31	0.46	1828.00
12/28/25	180.46	11.97	3143.97	3969.00	5775.21	19.54	158.86	21.60	0.46	1821.00

8. Body Composition Changes

Body Fat Category	Body Fat %	Date Reached	Description	Est. Time to Six-Pack
Above Average	20.00%	06/29/25	No visible abs, excess fat	4-6 months
Average	19.68%	07/06/25	Little muscle definition, soft look	3-4 months
Average	19.35%	07/13/25	Little muscle definition, soft look	3-4 months
Average	19.03%	07/20/25	Little muscle definition, soft look	3-4 months
Average	18.71%	07/27/25	Little muscle definition, soft look	3-4 months
Average	18.39%	08/03/25	Little muscle definition, soft look	3-4 months
Average	18.07%	08/10/25	Little muscle definition, soft look	3-4 months
Average	17.75%	08/17/25	Little muscle definition, soft look	3-4 months
Average	17.43%	08/24/25	Little muscle definition, soft look	3-4 months
Average	17.11%	08/31/25	Little muscle definition, soft look	3-4 months
Average	16.80%	09/07/25	Little muscle definition, soft look	3-4 months
Average	16.48%	09/14/25	Little muscle definition, soft look	3-4 months
Average	16.17%	09/21/25	Little muscle definition, soft look	3-4 months
Average	15.85%	09/28/25	Little muscle definition, soft look	3-4 months
Average	15.54%	10/05/25	Little muscle definition, soft look	3-4 months
Average	15.23%	10/12/25	Little muscle definition, soft look	3-4 months
Fit	14.91%	10/19/25	Some muscle definition, minimal fat	2-3 months
Fit	14.60%	10/26/25	Some muscle definition, minimal fat	2-3 months
Fit	14.30%	11/02/25	Some muscle definition, minimal fat	2-3 months
Fit	13.99%	11/09/25	Some muscle definition, minimal fat	2-3 months
Fit	13.68%	11/16/25	Some muscle definition, minimal fat	2-3 months
Fit	13.38%	11/23/25	Some muscle definition, minimal fat	2-3 months
Fit	13.08%	11/30/25	Some muscle definition, minimal fat	2-3 months
Fit	12.79%	12/07/25	Some muscle definition, minimal fat	2-3 months
Fit	12.50%	12/14/25	Some muscle definition, minimal fat	2-3 months
Fit	12.22%	12/21/25	Some muscle definition, minimal fat	2-3 months
Athletic	11.97%	12/28/25	Good muscle definition, low fat	1-2 months

9. Expected Results

Total Weight Loss:	19.54 lbs
Fat Loss:	8.03%
Muscle Gain:	-1.14 lbs

Average Weekly Changes:

Weight Loss:	0.75 lbs
Muscle Gain:	-0.04 lbs

Understanding Your Results:

- **Total Weight Loss:** Your projected loss of 19.54 lbs is designed to be sustainable and preserve muscle mass.
- **Fat Loss:** A reduction of 8.03% body fat represents a healthy rate of fat loss.
- **Muscle Changes:** The program aims to gain/maintain -1.14 lbs of muscle while losing fat.
- **Weekly Progress:** Targeting 0.75 lbs per week ensures steady, sustainable progress.

10. Metabolic Adaptation Forecast

Initial Week:	1.00 multiplier
Final Week:	0.96 multiplier
Expected Adaptation:	4.16% change
Lean Mass Preservation:	99.29%

Understanding Metabolic Adaptation:

- **Initial Response:** Your metabolism starts at 1.00x baseline, reflecting your current activity level.
- **Progressive Changes:** A 4.16% adaptation is expected as your body becomes more efficient.
- **Muscle Preservation:** The program is designed to maintain 99.29% of your lean mass.

11. Final Phase Targets

Maintenance Calories:	3969.00 calories/day
Weekly Energy Output:	5775.21 calories

Understanding Your Final Targets:

- **Maintenance Level:** Your final daily maintenance of 3969.00 calories reflects your new metabolic baseline.
- **Weekly Output:** Maintaining a weekly energy output of 5775.21 calories will help sustain your results.
- **Long-term Success:** These targets are designed to help you maintain your new weight and body composition.

12. Input Parameter Analysis

All Input Parameters Used in Calculations:

This section confirms that all provided input parameters are directly used in the body composition calculations.

Parameter	Value	Used in Calculation	Calculation Component
Athlete Status	n	✓ Yes	RMR boost (10% increase)
Bodybuilder Mode	True	✓ Yes	Calorie limits and muscle gain adjustments
Current Body Fat	20.0	✓ Yes	Body composition calculations
Current Weight	200.0	✓ Yes	RMR calculation (Mifflin-St Jeor), TDEE, muscle gain estimation
Date of Birth	01/01/94	✓ Yes	Age calculation for RMR and muscle gain adjustments
Diet Type	high_protein	✓ Yes	Muscle gain and fat loss multipliers
Exercise Type	resistance	✓ Yes	NEAT boost and muscle gain adjustments
Gender	m	✓ Yes	RMR calculation (different formulas for male/female)
Goal Body Fat	12.0	✓ Yes	Dual-goal deficit calculations
Goal Weight	180.0	✓ Yes	Dual-goal deficit calculations
Height (Feet)	6	✓ Yes	RMR calculation (converted to height_cm)
Height (Inches)	0	✓ Yes	RMR calculation (converted to height_cm)
Job Activity	2	✓ Yes	NEAT estimation (100-700 calories)
Leisure Activity	2	✓ Yes	NEAT estimation (50-350 calories)
PED Use	False	✓ Yes	Muscle gain multiplier (1.5x) and fat loss ratio
Protein Intake	180.0	✓ Yes	TEF calculation (25% thermic effect) and muscle gain
Resistance Training	y	✓ Yes	Muscle gain calculation (0.05 baseline if False, full calculation if True)
Sleep Quality	good	✓ Yes	Muscle gain multiplier (1.0 vs 0.8)
Training Experience	Intermediate (2-4 years)	✓ Yes	Base muscle gain rates (0.0025-0.0125 weekly)

Input Reliability Analysis:

Category	Score	Assessment
Anthropometric Data	100.0/100	Excellent
Activity Data	100.0/100	Excellent
Goal Data	100.0/100	Excellent
Data Completeness	70.0/100	Mostly Complete

13. AI Confidence Analysis

AI-Powered Assessment of Your Plan:

This analysis uses advanced AI to evaluate the reliability and feasibility of your body composition goals.

Analysis Component	Score	Assessment
Overall Confidence	75.0/100	High
Input Reliability	85.0/100	Very Reliable
Calculation Accuracy	90.0/100	Very Accurate
Goal Feasibility	80.0/100	Feasible

Score Explanations:

Component	Score	AI Explanation
Overall Confidence	75.0/100	The plan shows good potential for success with moderate confidence due to reasonable calorie deficit and body composition goals. However, the missing timeframe data and moderate data completeness score impact overall confidence.
Input Reliability	85.0/100	High reliability scores for anthropometric, activity, and goal data (100/100 each) indicate accurate baseline measurements. The lower data completeness score (70/100) and missing timeframe affect overall input reliability.
Calculation Accuracy	90.0/100	TDEE and calorie calculations appear precise and align with established formulas. The targeted weekly weight loss of 0.72 lbs represents a sustainable 370-calorie daily deficit, which is well within evidence-based recommendations.
Goal Feasibility	80.0/100	The goal to lose 20 lbs while reducing body fat from 20% to 12% is achievable. The moderate calorie deficit suggests minimal muscle loss risk, supporting the body composition goals.

⚠ Important Warnings:

- Missing timeframe data could impact goal setting and progress tracking
- TDEE seems relatively high at 4091 calories - verify activity level accuracy
- Body fat reduction of 8% may require longer-term commitment

AI Recommendations:

- Define specific timeframe for goal achievement
- Implement weekly body composition measurements
- Consider breaking goal into 4-week milestone targets
- Verify TDEE through 2-week calorie tracking
- Add resistance training protocol to preserve muscle mass

Detailed Confidence Factors:

Factor	Score
Timeframe Realism	60.0/100
Calorie Deficit Sustainability	85.0/100
Body Composition Feasibility	80.0/100
Data Quality	85.0/100

Detailed AI Analysis Summary:

The transformation plan demonstrates good potential for success with some notable considerations. The calculated daily deficit of 370 calories should yield sustainable fat loss while minimizing muscle loss risk. The goal of reducing body fat from 20% to 12% while losing 20 lbs total weight is physiologically feasible, especially given the moderate approach to calorie restriction. The high TDEE of 4091 calories warrants verification to ensure accuracy of the deficit calculations. The missing timeframe data represents the most significant limitation in the plan, as it affects the ability to set appropriate progression milestones and adjust strategies as needed. The high quality of anthropometric and activity data provides a solid foundation for the plan, though the 70/100 data completeness score suggests room for additional detail. To optimize success probability, implementing the suggested improvements - particularly defining a specific timeframe and adding regular body composition tracking - would be beneficial. The overall 75% confidence score reflects good fundamental planning with some areas requiring refinement for optimal results.

14. Comprehensive Parameter Reference Guide

This section details exactly how each input parameter affects your calculations and results:

Parameter Usage in Calculations:

Parameter	Your Value	Usage Location	Impact on Results
Age	30 years	RMR Calculation (Mifflin-St Jeor)	Base metabolic rate decreases ~1-2% per decade after age 30
Weight	200.0 lbs	RMR, TDEE, Body Composition	Primary factor in all metabolic calculations - higher weight = higher energy needs
Height	6'0" (182.88 cm)	RMR Calculation, BMI	Taller individuals have higher metabolic rates due to larger body surface area
Gender	m	RMR Calculation (different formulas)	Males typically have 10-15% higher RMR than females due to muscle mass differences
Activity Level	Light exercise/sports 1-3 days/week	TDEE Multiplier	Multiplies RMR by 1.2-1.9 based on activity level
Body Fat %	20.0%	Lean Body Mass, Metabolic Calculations	Lower body fat = higher lean mass = higher metabolic rate per pound
Goal Weight	180.0 lbs	Calorie Adjustment, Macro Distribution	Target weight determines caloric deficit/surplus needed
Resistance Training	Yes	PRIME_Diet_Calculations_v2.py:161	Increases protein needs and metabolic boost
Athlete Status	Yes	PRIME_Utills.py:105-106	Adds 10% RMR boost for enhanced metabolic capacity
Workout Type	Bodybuilding	Macro Distribution, TEF Calculations	Bodybuilding training affects protein/carb/fat ratios and thermic effect
Experience Level	Intermediate (2-4 years)	NEAT Estimation, Performance Adjustments	Intermediate (2-4 years) affects recovery needs and energy expenditure patterns
PED Use	No	PRIME_Calculations.py, PRIME_Diet_Calculations_v2.py	Standard natural muscle gain and fat loss rates apply

Calculation Impact Summary:

- **Your RMR:** 1900 calories - calculated using age (30), weight (200.0lbs), height (6'0" (182.88 cm)), and gender (m)
- **Your TDEE:** 4091 calories - RMR multiplied by activity factor for Light exercise/sports 1-3 days/week activity level
- **Your Goal Calories:** 3721 calories - TDEE adjusted for your goal
- **Body Fat Impact:** 20.0% body fat indicates 160lbs lean mass, affecting metabolic calculations
- **Athlete Bonus:** 10% RMR increase applied due to athlete status

Code References:

- **Resistance Training Check:** PRIME_Diet_Calculations_v2.py line 161 - Uses 0.05 baseline if False
- **Athlete RMR Boost:** PRIME_Utils.py lines 105-106 - Adds 10% if True
- **Diet Type Multipliers:** Multiple locations with fat loss/muscle gain adjustments
- **Exercise Type NEAT:** NEAT estimation and performance adjustment calculations
- **Activity Level Multipliers:** Standard TDEE calculation with activity-specific factors
- **PED Use Multipliers:** PRIME_Calculations.py line 81 - 1.5x muscle gain;
PRIME_Diet_Calculations_v2.py lines 109-110, 166-167 - Enhanced fat loss and muscle gain

14. Comprehensive Reference Guide

Activity Levels and Caloric Impact

Level	Description	TDEE Multiplier	Daily Caloric Impact	Example Activities
Sedentary	Little to no exercise, desk job	1.2	+20% above RMR	Office work, driving, watching TV
Lightly Active	Light exercise 1-3 days/week	1.375	+37.5% above RMR	Walking, light gardening, yoga
Moderately Active	Moderate exercise 3-5 days/week	1.55	+55% above RMR	Jogging, cycling, recreational sports
Very Active	Heavy exercise 6-7 days/week	1.725	+72.5% above RMR	Construction work, competitive sports
Extremely Active	Very heavy exercise, physical job	1.9	+90% above RMR	Professional athletes, heavy manual labor

Training Experience and Muscle Gain Potential

Level	Years of Training	Monthly Muscle Gain	Caloric Requirements	Recovery Needs
Beginner	0-2 years	0.5-1% body weight	+500 cal/day surplus	24-36 hours between sessions
Intermediate	2-4 years	0.25-0.5% body weight	+300-400 cal/day surplus	36-48 hours between sessions
Advanced	4+ years	0.125-0.25% body weight	+200-300 cal/day surplus	48-72 hours between sessions

Workout Types and Metabolic Impact

Type	Primary Focus	Caloric Burn	EPOC Effect	Recovery Demand
Bodybuilding	Muscle growth and aesthetics	600-800 cal/session	24-36 hours	High protein needs (1g/lb)
Strength Training	Maximal strength	400-600 cal/session	24-48 hours	High CNS recovery needs
Powerlifting	Specific lift performance	500-700 cal/session	36-48 hours	Very high CNS demands
Olympic Weightlifting	Power and technique	600-800 cal/session	24-36 hours	Technical recovery focus
CrossFit	Overall fitness	800-1000 cal/session	36-48 hours	High systemic demands
Calisthenics	Bodyweight control	400-600 cal/session	24-36 hours	Joint/tendon focus

Resistance Training Impact

Status	Muscle Preservation	Metabolic Impact
With RT	90-100%	+100-300 cal/day
Without RT	60-80%	No additional benefit

Impact on Calculations: Resistance training increases your daily caloric needs by 100-300 calories through increased muscle mass and EPOC (Excess Post-exercise Oxygen Consumption).

Athlete Status Effects

Status	Metabolic Rate	Nutrient Partitioning
Athlete	+10-15% higher	Enhanced efficiency
Non-Athlete	Standard	Normal efficiency

Impact on Calculations: Athletic status can increase your TDEE by 10-15% due to enhanced metabolic efficiency and muscle mass.

Volume Score Components

Score Range	Sets per Muscle	Weekly Volume
0.0-0.2	1-5 sets	Low stimulus
0.2-0.4	6-10 sets	Maintenance
0.4-0.6	10-15 sets	Growth stimulus
0.6-0.8	15-20 sets	Optimal growth
0.8-1.0	20+ sets	Maximum stimulus

Intensity Score Impact

Score Range	% of 1RM	Rep Range
0.0-0.2	30-50%	15+ reps
0.2-0.4	50-65%	12-15 reps
0.4-0.6	65-75%	8-12 reps
0.6-0.8	75-85%	5-8 reps
0.8-1.0	85%+	1-5 reps

Frequency Score Effects

Score Range	Sessions/Week	Recovery Time
0.0-0.2	1x/week	7 days
0.2-0.4	2x/week	3-4 days
0.4-0.6	3x/week	2-3 days
0.6-0.8	4x/week	1-2 days
0.8-1.0	5+x/week	<1 day

Calculation Formulas and Methodology

Component	Formula	Variables	Example
RMR (Male)	$(10 \times \text{weight kg}) + (6.25 \times \text{height cm}) - (5 \times \text{age}) + 5$	Weight, Height, Age	70kg man: 1,745 cal
RMR (Female)	$(10 \times \text{weight kg}) + (6.25 \times \text{height cm}) - (5 \times \text{age}) - 161$	Weight, Height, Age	60kg woman: 1,374 cal
TDEE	RMR $\times$ Activity Multiplier + Exercise Calories	RMR, Activity Level	$1,745 \times 1.55 = 2,705$ cal
Caloric Deficit	TDEE - Daily Intake	TDEE, Target Loss	$2,705 - 2,205 = 500$ cal
Weekly Fat Loss	$(\text{Weekly Deficit}) \div 3,500$	Daily Deficit $\times 7$	$(500 \times 7) \div 3,500 = 1$ lb
Muscle Gain	$(\text{Body Weight} \times \text{Experience Factor})$	Weight, Training Level	$70\text{kg} \times 0.01 = 0.7\text{kg/month}$